SAMPLE

Earth's Sky: The Milky Way from Earth

How to See Our Own Galaxy

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Introduction

The one galaxy this book is about is the only one the reader will ever be inside of. The Milky Way, the pale band that crosses the whole sky on a moonless night far from city light, is our own galaxy seen edge-on from within, a hundred billion suns' worth of combined glow, spread across the heavens by the accident of our address: we live in the disk, and the disk, seen from inside, surrounds us as a ring. Every other galaxy in this catalog's volumes is a distant object, seen whole from outside; this one is our house, and the reader who learns to read its band learns the layout of the house.

The volume follows the series' standing method: the sky's phenomena treated as working knowledge for the observer, with the mechanics behind them, the instruments and history that sorted them out, and the honest accounting of what remains difficult. For this subject the difficulty is unusual and worth stating at the start: we cannot step outside to look at our galaxy, so everything in the band, its shape, its size, our place in it, had to be deduced from inside, a problem the astronomers compare to mapping a city from one street corner, and the solution of that problem across four centuries is one of astronomy's great stories, told in the chapters ahead.

The preview for the reader in a hurry: the band is the galaxy's disk in profile; its light is the summed glow of distant suns too far apart to separate by eye; its dark lanes are dust clouds in front of them; its brightest stretch, in the southern summer sky, points toward the galaxy's hidden center; the whole structure was worked out over centuries, from the first telescopic resolution of the band into stars, through the star-counting that mapped the disk, to the great debate about the galaxy's size and our place in it; and in the modern era the band has become a heritage question, because the majority of the world's people now live where light pollution has erased it, and seeing our own galaxy has become something that must be traveled for. The chapters unpack each clause, and the closing chapters say where to go and when.

Chapter 1: The Discovery of The Milky Way from Earth

The discovery story of this subject is the story of a band of light that every human who ever lived under a dark sky saw, and that only the last four hundred years of thought has explained, and the chapters tell it as a relay of insights rather than a single eureka.

The ancient watching comes first, and it deserves better than the summary histories give it. Every culture under a dark sky named the band, and the names are a catalog of human imagination: the milk of a goddess in the Greek and Latin worlds, whose name the modern term galaxy still carries; a river of heaven in the Chinese tradition, a celestial water with banks and fords; the road of the dead or of the gods among the northern peoples; a spine in the sky, a serpent, a tree, a seam along which the heavens were sewn. The Inca, watching from their mountain valleys, saw a river in the band and dark animals in its patches; the desert peoples tracked its seasonal leanings; the navigators of the southern oceans knew its dark patches as landmarks. The reader should take the ancient record seriously as observation: the names distinguish the bright stretches, the dark gaps, the seasonal changes, and the southern richness, which means the old watchers had done the first, hardest part of the science, they had looked carefully and remembered what they saw.

A note on method belongs in the ancient record too, because the old watchers' lore was working astronomy, not only poetry: the Polynesian navigators steered by the band's rising points as a compass of their ocean crossings; the Inca's dark animals marked the seasonal stations of the husbandry year; the Chinese river's fords and banks catalogued features the modern charts still use as signposts. The band was humanity's first reference frame, and the science of this book is, in the honest sense of the phrase, a very old project.

The instrumental discovery began in the winter of 1610, when the telescope's first great user turned his little instrument on the band and found, in his own delighted words, that it was nothing but a mass of innumerable stars, packed so close that they mingled their light. The band's mystery, whether it was cloud or vapor or road, ended in one evening: it was stars, beyond counting, at distances the eye could not divide. The discovery belongs to this book as its foundation, because everything after is the working out of what a band of unresolved starlight means about the arrangement of the stars around us.

The shape came next, and slowly. An eighteenth-century musician-turned-astronomer, sweeping the sky with his home-built telescopes and counting stars in every direction, found the band was a plane: counts along it ran high in a long circle of the sky, counts away from it fell off quickly, and he sketched the result as a disk of stars with the sun somewhere near its middle, the so-called grindstone model. The sketch was wrong about our address, but

right about the geometry, and the method, counting stars to infer structure, founded the science the ninth chapter describes. A philosopher of the same era, reading of the nebulae, had already proposed that the band was itself a vast wheel of stars, one among many such islands, a guess the next chapter's history vindicated.

Our true address arrived in the twentieth century's opening years. A young American astronomer, counting and timing a class of compact, spherical star clusters that swarms thickly in the band's southern half, found their center of distribution not in the band's brightest part near us but far off in the direction of the constellation of the archer, and concluded the sun was far from the galaxy's center, tens of thousands of light-years out in the disk. The adjustment was the largest any human had made to their own address since the old argument about the Earth's place among the planets, and the debates it triggered, culminating in the famous two-astronomer showdown of 1920 over the galaxy's size and whether the nebulae were inside it, belong to the seventh chapter's context. The discovery chapter's summary: everyone saw the band; the telescope showed it was stars; the counters showed it was a disk; the clusters showed we are not at its center; and the modern reader, inheriting all four steps, looks at the band knowing what four centuries of effort it took to see it plainly.

A paragraph on the oldest right answer in the file, the guess that waited two thousand years for its witness. Among the ancient speculations about the sky's pale river, one school of Greek atomists proposed the true explanation: the band, they suggested, was not a celestial substance at all but a crowd of stars, packed so densely that their separate lights blended into a continuous glow to the unaided eye; the claim was made without instruments and was, by the standards of the time, untestable, a philosopher's hunch filed among better-funded errors. The origin note for the file: the hunch was vindicated when the first telescopes turned on the band and dissolved the milk into stars, the ancient guess confirmed in a single evening, and the file's lesson is about the two kinds of correctness, the lucky guess and the tested claim, the atomists having supplied the first and the telescope the second, the sky's brightest riddle having been answered in words centuries before it was answered in evidence.

A second origin note, on the evening the river broke, the telescope's conversion of the glow into a census. The first telescopic observers, sweeping the band, reported the same shock: the pale river, smooth to the eye, shattered into swarms of individual stars, the eyepiece resolving what the eye had blended, every field of the band a street of suns; the observation, recorded and published in the early telescope age, settled the ancient debate and established the band as a depth problem, the glow being distance, the smoothness being crowding, the Milky Way becoming the first object whose appearance depended on the limitations of the observer. The origin's closing note: the file should keep the shock of that first view alive, because it is the founding experience of the subject, every later reader with a telescope repeating it, and the band remains the standard test of an instrument's reach, the sky's own resolution chart, the place where the telescope proves that the smooth things of the world are crowds, and that distance turns crowds into texture.